graphify-plus v2.1

How to Run Guide

Complete setup guide for running graphify-plus on Windows.
Covers CMD, PowerShell, Claude Code, and VS Code integration.
Includes Ollama semantic enrichment and all 14 MCP tools.

Machine	`Windows`
Python	`3.11`
Proxy	`192.168.15.53:3128`
Venv	`C:\Users\Gunju Kumari\Documents\Claude\Projects\venv311\`
Project	`C:\Users\Gunju Kumari\Documents\Claude\Projects\graphify\`
Database	`C:\Users\Gunju Kumari\.graphify-plus\graph.db`

Method 1: PowerShell (Recommended)

Step 1: Open PowerShell and Activate Venv

Open PowerShell (search "PowerShell" in Start menu), then run:

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned
& "C:\Users\Gunju
Kumari\Documents\Claude\Projects\venv311\Scripts\Activate.ps1"
```

You should see `(venv311)` at the start of your prompt.

Step 2: Navigate and Set PYTHONPATH

```
cd "C:\Users\Gunju Kumari\Documents\Claude\Projects\graphify"
$env:PYTHONPATH = "src"
```

IMPORTANT: In PowerShell, use `$env:PYTHONPATH = "src"`. Do NOT use `set PYTHONPATH=src` (that is CMD syntax and does nothing in PowerShell).

Step 3: Start the Server

```
python -m graphify_plus.server
```

You should see:

```
graphify-plus v0.2.0 running at http://127.0.0.1:3000
Dashboard: http://127.0.0.1:3000
```

Open `http://localhost:3000` in your browser.

Step 4: Quick Copy-Paste Block (all in one)

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned
& "C:\Users\Gunju
Kumari\Documents\Claude\Projects\venv311\Scripts\Activate.ps1"
cd "C:\Users\Gunju Kumari\Documents\Claude\Projects\graphify"
$env:PYTHONPATH = "src"
python -m graphify_plus.server
```

Method 2: Command Prompt (CMD)

Step 1: Open CMD and Activate Venv

Open CMD (search "cmd" in Start menu), then run:

```
"C:\Users\Gunju Kumari\Documents\Claude\Projects\venv311\Scripts\activate.bat"
```

Step 2: Navigate and Set PYTHONPATH

```
cd "C:\Users\Gunju Kumari\Documents\Claude\Projects\graphify"
set PYTHONPATH=src
```

NOTE: In CMD, use `set PYTHONPATH=src` (no quotes, no $env:). This is the opposite of PowerShell!

Step 3: Start the Server

```
python -m graphify_plus.server
```

Quick Copy-Paste Block (CMD)

```
"C:\Users\Gunju Kumari\Documents\Claude\Projects\venv311\Scripts\activate.bat"
cd "C:\Users\Gunju Kumari\Documents\Claude\Projects\graphify"
set PYTHONPATH=src
python -m graphify_plus.server
```

Method 3: One-Click Batch File (Easiest)

Create a file called `start-graphify.bat` on your Desktop with this content:

```
@echo off
echo Starting graphify-plus...
call "C:\Users\Gunju
Kumari\Documents\Claude\Projects\venv311\Scripts\activate.bat"
cd /d "C:\Users\Gunju Kumari\Documents\Claude\Projects\graphify"
set PYTHONPATH=src
start http://localhost:3000
python -m graphify_plus.server
pause
```

Double-click this .bat file to start graphify-plus. It activates the venv, opens the browser, and starts the server automatically.

How to Create the .bat File

1. Right-click on your Desktop > New > Text Document
2. Paste the code above
3. File > Save As > change "Save as type" to "All Files"
4. Name it `start-graphify.bat`
5. Click Save
6. Double-click the .bat file to run

Ollama Setup (FREE Semantic Enrichment)

Install Ollama

1. Go to https://ollama.com
2. Download for Windows
3. Run the installer (it starts automatically in background)

Pull a Model (with Proxy)

Open a NEW PowerShell window:

```
$env:HTTPS_PROXY = "http://192.168.15.53:3128"
$env:HTTP_PROXY = "http://192.168.15.53:3128"
ollama pull phi3
```

Or in CMD:

```
set HTTPS_PROXY=http://192.168.15.53:3128
set HTTP_PROXY=http://192.168.15.53:3128
ollama pull phi3
```

Recommended Models

Model	Size	Speed	Best For
`phi3`	2.3 GB	Fast (CPU OK)	Light machines, quick results
`codellama`	3.8 GB	Medium	Code-optimized
`llama3.1`	4.7 GB	Slow on CPU	Best quality (needs GPU)
`qwen2:1.5b`	1.0 GB	Very Fast	Weakest machine

> If `llama3.1` times out, use `phi3` instead. It runs on any machine without a GPU.

Run Semantic Enrichment

1. Start graphify-plus server (see Method 1, 2, or 3 above)
2. Open http://localhost:3000
3. Click **"Enrich with AI"** on your project card
4. Select model from dropdown (e.g., `phi3:latest`)
5. Click **"Enrich with Ollama (FREE)"**
6. Wait for progress bar to complete

Claude Code MCP Setup

Step 1: Install Plugin from Dashboard

In the dashboard at `http://localhost:3000`:

1. Click **"Install Plugin"** on your project card
2. Select **"Claude Code"**
3. This creates a `.mcp.json` file in your project root

Step 2: Start Claude Code

Open a NEW PowerShell window:

```
$env:HTTPS_PROXY = "http://192.168.15.53:3128"
$env:HTTP_PROXY = "http://192.168.15.53:3128"
cd "C:\Users\Gunju Kumari\Desktop\audio_player_gunju"
npx @anthropic-ai/claude-code
```

Replace `audio_player_gunju` with your actual project folder path.

Step 3: Verify MCP Tools

In Claude Code, type:

```
> What MCP tools do you have?
```

You should see 14 tools listed.

Step 4: Token-Saving Workflow

Ask questions in this order for minimum token usage:

```
# 1. Understand the codebase structure (~300 tokens)
> Show me the community summaries

# 2. Search by meaning (~200 tokens)
> Search the graph for "audio playback"

# 3. Check file before reading it (~80 tokens)
> Get file summary for app.py

# 4. Only read source when needed (~2000 tokens)
> Now read app.py, I need the play function code
```

VS Code + Cline Setup

1. In the dashboard, click **"Install Plugin"** > **"VS Code Cline"**
2. This creates `.mcp.json` in your project root
3. Open the project folder in VS Code
4. Cline extension auto-detects the MCP server
5. You can now ask Cline to use graph tools

All 14 MCP Tools Reference

#	Tool	What it Does	Tokens
1	`list_projects`	List all indexed projects	~50
2	`graph_stats`	Node/edge/community counts	~30
3	`search_graph`	Find nodes by meaning (FTS5 + semantic)	50-200
4	`get_god_nodes`	Most connected hub nodes	100-200
5	`get_neighbors`	Connections around a node	100-500
6	`shortest_path`	How two concepts connect	200-500
7	`nodes_in_file`	All nodes from a file	50-500
8	`get_community`	All nodes in a cluster	100-1000
9	`rebuild_graph`	Rebuild after code changes	0
10	`diff_impact`	Git changes > graph impact	100-500
11	`semantic_enrich`	Run Ollama enrichment	0
12	`get_file_summary`	Describe file without reading source	50-100
13	`community_summaries`	Table-of-contents for codebase	100-500

Token Savings Summary

Approach	Tokens/Query	Savings
No graph (AI reads all files)	~50,000	0% (baseline)
GRAPH_REPORT.md only	~2,000	96%
MCP tools (before semantic)	~500	99%
MCP tools (after semantic)	~200	99.6%

The 7 Token-Saving Strategies

Strategy	What	Savings
1. Semantic Enrichment (Ollama)	Search by meaning, not name. FREE.	~30% fewer file reads
2. Compact Mode	search_graph(compact=true) strips extra fields	~40% per search
3. Enhanced GRAPH_REPORT.md	Report includes descriptions + concepts	~80% vs reading files
4. Community Summaries	Skip irrelevant clusters entirely	Huge on large projects
5. get_file_summary	Decide if you need source without reading it	~2000 tok per file skip
6. FTS5 Search	Instant prefix/substring matching	Faster queries
7. Incremental Rebuild	Preserves semantic data on file save	No re-enrichment

Troubleshooting

Problem: `ModuleNotFoundError: No module named 'graphify_plus'`

Cause: You forgot to set PYTHONPATH.

Fix:

```
PowerShell: $env:PYTHONPATH = "src"
CMD: set PYTHONPATH=src
```

Problem: `Server exits silently (no output)`

Cause: Old server.py without __main__ block.

Fix:

```
Download the latest server.py from Claude.
```

Problem: `'C:\Users\Gunju' is not recognized`

Cause: Path has spaces and wasn't quoted.

Fix:

```
Use: python -m graphify_plus.server
Do NOT run the .py file directly.
```

Problem: `Ollama: dial tcp: lookup registry.ollama.ai`

Cause: Proxy not set for Ollama.

Fix:

```
$env:HTTPS_PROXY = "http://192.168.15.53:3128"
ollama pull phi3
```

Problem: `Ollama: timed out`

Cause: Model too large for your CPU.

Fix:

```
Use phi3 (2.3GB) or qwen2:1.5b (1GB) instead of llama3.1.
```

Problem: `MCP tools not showing in Claude Code`

Cause: .mcp.json missing or wrong path.

Fix:

```
Click "Install Plugin" > "Claude Code" in dashboard.
```

Problem: `Fresh start / reset database`

Cause: Delete the database file.

Fix:

```
Remove-Item "C:\Users\Gunju Kumari\.graphify-plus\graph.db"
```